



IMPACT CHALLENGE

Problem to Prototype

AI-Powered Cyber Threat Intelligence

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Objective

What the Project Aims to Achieve – To develop a simple and reliable Arduino-based smart control system that can automatically operate multiple LEDs according to a programmed sequence.

Main Goal of the IoT Solution – To demonstrate how smart automation and microcontroller technology can be used to control electronic devices efficiently with minimal human intervention.

Problem Statement

Real-World Application – To demonstrate a concept that can be adapted for smart lighting, traffic management, warning systems, security indicators, and other automated applications.



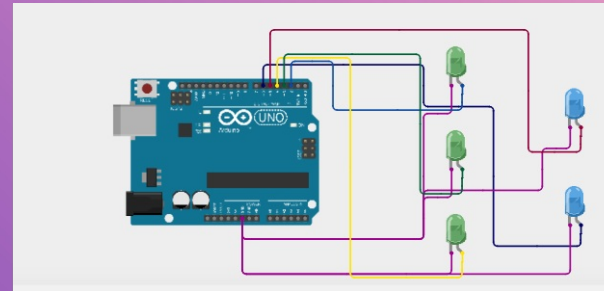
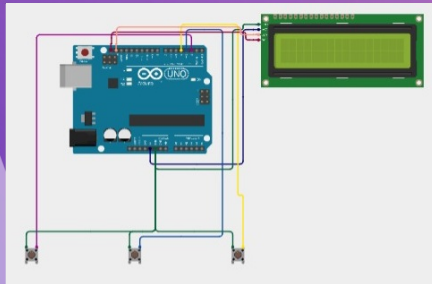
Proposed Solution & Circuit Diagram

How the Model Works

Three buttons simulate suspicious security events: unusual login, sensitive-file access, and USB data copying.

The Arduino records the events and checks whether they are linked to the same user and workstation and occur in the correct order within 10 minutes.

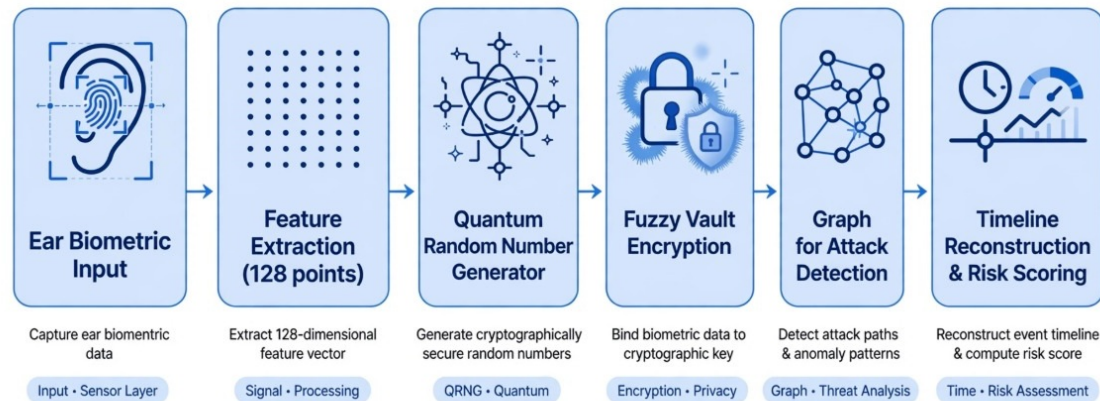
A rules-based risk score is then calculated. If all events are connected, the score reaches 100/100, and the LCD displays the detected incident and recommended actions.



Future Enhancement

Cybersecurity System Architecture Diagram

Ear Biometric Authentication → Risk Analysis Pipeline



Secure • Privacy-Preserving

Biometric data protected via fuzzy vault



Real-time • Threat Detection

Graph-based attack chain identification



Actionable • Risk Insights

Timeline + Risk score for response

Working Model Explanation

Our innovation is a physical-digital threat mesh for visualizing cyberattacks. Instead of showing security alerts only on a dashboard, our system represents different parts of a network as interconnected nodes. When events such as brute-force attacks, new tools, privilege escalation, or large data transfers occur, the threat propagates through the connected nodes and their threat levels change in real time.

We also demonstrate a zero-trust approach, where neighboring nodes share threat information and a compromised segment can be isolated. For this prototype, we implemented the concept as an interactive digital model, which can later be connected to physical microcontrollers and LEDs.

The innovation is that we don't just display individual alerts. We correlate them and visualize how a threat moves through the network, while allowing the affected segment to be isolated.

LIVE THREAT TOPOLOGY

INTERNET

WEB SERVER

• SAFE

APPLICATION

• SAFE

DATABASE

• SAFE

0

THREAT SCORE

1

EVENTS

0

AFFECTED NODES

ATTACK SIMULATION

⚠ BRUTE FORCE

Repeated authentication failures

⚠ NEW TOOL

Unknown tool detected

⚠ PRIVILEGE ESCALATION

Suspicious privilege increase

⚠ LARGE EGRESS

Abnormal outbound data transfer

🛡 ISOLATE DATABASE

Segment compromised node

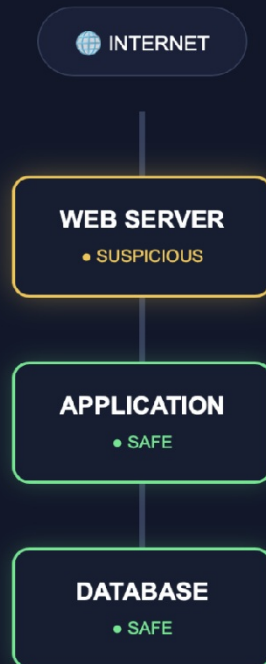
↺ RESET SYSTEM

Return to normal state

EVENT STREAM

02:52:28 PM VERTEX mesh reset and initialized.

LIVE THREAT TOPOLOGY



15

THREAT SCORE

2

EVENTS

1

AFFECTED NODES

ATTACK SIMULATION

⚠ BRUTE FORCE
Repeated authentication failures

⚠ NEW TOOL
Unknown tool detected

⚠ PRIVILEGE ESCALATION
Suspicious privilege increase

⚠ LARGE EGRESS
Abnormal outbound data transfer

🛡 ISOLATE DATABASE
Segment compromised node

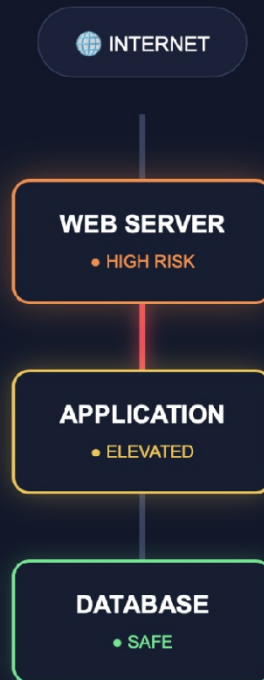
🔄 RESET SYSTEM
Return to normal state

EVENT STREAM

02:56:11 PM BRUTE_FORCE detected at WEB_NODE

02:52:28 PM VERTEX mesh reset and initialized.

LIVE THREAT TOPOLOGY



35

THREAT SCORE

4

EVENTS

2

AFFECTED NODES

ATTACK SIMULATION

⚠ BRUTE FORCE
Repeated authentication failures

⚠ NEW TOOL
Unknown tool detected

⚠ PRIVILEGE ESCALATION
Suspicious privilege increase

⚠ LARGE EGRESS
Abnormal outbound data transfer

🛡 ISOLATE DATABASE
Segment compromised node

🔄 RESET SYSTEM
Return to normal state

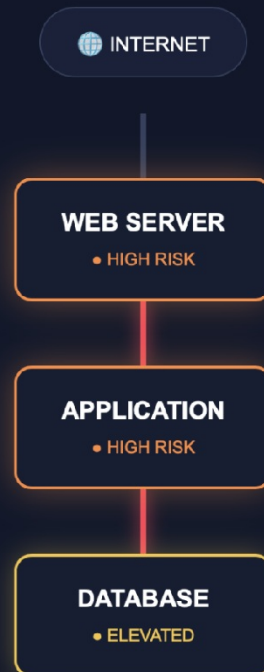
EVENT STREAM

02:57:15 PM GOSSIP → APP_NODE: anomaly propagated

02:57:14 PM NEW_TOOL detected at WEB_NODE

02:56:11 PM BRUTE_FORCE detected at WEB_NODE

LIVE THREAT TOPOLOGY



60

THREAT SCORE

6

EVENTS

3

AFFECTED NODES

ATTACK SIMULATION

⚠ BRUTE FORCE

Repeated authentication failures

⚠ NEW TOOL

Unknown tool detected

⚠ PRIVILEGE ESCALATION

Suspicious privilege increase

⚠ LARGE EGRESS

Abnormal outbound data transfer

🛡 ISOLATE DATABASE

Segment compromised node

🔄 RESET SYSTEM

Return to normal state

EVENT STREAM

02:58:16 PM GOSSIP → DATABASE_NODE: threat propagated

02:58:15 PM PRIVILEGE_ESCALATION detected at APP_NODE

02:57:15 PM GOSSIP → APP_NODE: anomaly

LIVE THREAT TOPOLOGY



125

THREAT SCORE

10

EVENTS

3

AFFECTED NODES

ATTACK SIMULATION

⚠ BRUTE FORCE

Repeated authentication failures

⚠ NEW TOOL

Unknown tool detected

⚠ PRIVILEGE ESCALATION

Suspicious privilege increase

⚠ LARGE EGRESS

Abnormal outbound data transfer

🛡 ISOLATE DATABASE

Segment compromised node

🔄 RESET SYSTEM

Return to normal state

EVENT STREAM

02:59:15 PM DATA EXFILTRATION suspected

02:59:15 PM LARGE_EGRESS detected at
DATABASE_NODE

02:59:14 PM GOSSIP → DATABASE_NODE: threat
propagated

LIVE THREAT TOPOLOGY

INTERNET

WEB SERVER

• SAFE

APPLICATION

• SAFE

DATABASE

• CRITICAL

40

THREAT SCORE

4

EVENTS

1

AFFECTED NODES

✓ DATABASE SEGMENT ISOLATED

ATTACK SIMULATION

⚠ BRUTE FORCE

Repeated authentication failures

⚠ NEW TOOL

Unknown tool detected

⚠ PRIVILEGE ESCALATION

Suspicious privilege increase

⚠ LARGE EGRESS

Abnormal outbound data transfer

🛡 ISOLATE DATABASE

Segment compromised node

🔄 RESET SYSTEM

Return to normal state

EVENT STREAM

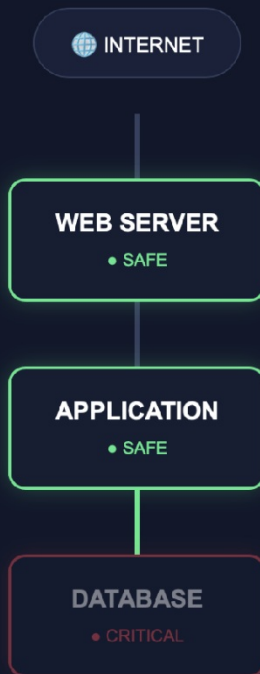
03:00:16 PM ZERO-TRUST ACTION: DATABASE NODE ISOLATED

03:00:14 PM DATA EXFILTRATION suspected

03:00:14 PM LARGE_EGRESS detected at DATABASE_NODE

03:00:12 PM VERTEX mesh reset and initialized

LIVE THREAT TOPOLOGY



40

THREAT SCORE

5

EVENTS

1

AFFECTED NODES

✓ DATABASE SEGMENT ISOLATED

ATTACK SIMULATION

⚠ BRUTE FORCE
Repeated authentication failures

⚠ NEW TOOL
Unknown tool detected

⚠ PRIVILEGE ESCALATION
Suspicious privilege increase

⚠ LARGE EGRESS
Abnormal outbound data transfer

🛡 ISOLATE DATABASE
Segment compromised node

🔄 RESET SYSTEM
Return to normal state

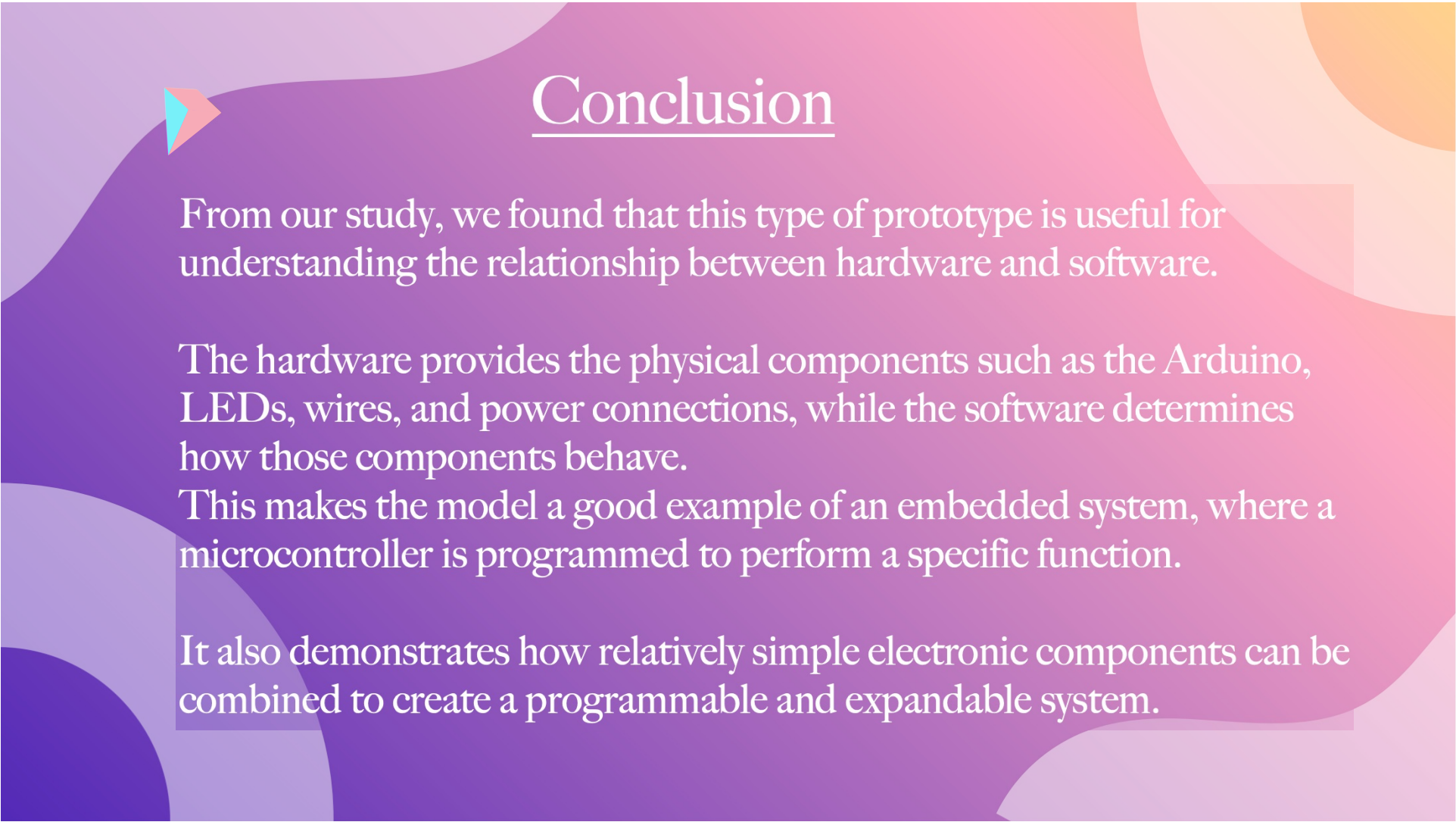
EVENT STREAM

03:00:54 PM ZERO-TRUST ACTION: DATABASE NODE ISOLATED

03:00:16 PM ZERO-TRUST ACTION: DATABASE NODE ISOLATED

03:00:14 PM DATA EXFILTRATION suspected

03:00:14 PM LARGE EGRESS detected at



Conclusion

From our study, we found that this type of prototype is useful for understanding the relationship between hardware and software.

The hardware provides the physical components such as the Arduino, LEDs, wires, and power connections, while the software determines how those components behave.

This makes the model a good example of an embedded system, where a microcontroller is programmed to perform a specific function.

It also demonstrates how relatively simple electronic components can be combined to create a programmable and expandable system.